

Ocean Colour and Chlorophyll Concentration As a Proxy for Phytoplankton Biomass Along the South-West Coast of England During the Coccolithophore Spring Blooms

1. Introduction

Phytoplankton blooms are a seasonal occurrence in the South-Western English Channel and are driven by an increase in sunlight during spring (March to May), nutrient input from river runoff and rainfall as well as wind-driven vertical mixing due to winter storms (*Western English Channel Springs Into Life*, 2021). The combination of these factors creates an ideal environment for phytoplankton to photosynthesise in the well-mixed, high-nutrient, shallow, sunlit surface layers where they grow in size and in population. This can be observed remotely by using chlorophyll which is found within all photosynthetic organisms.

Within the South-Western English Channel, dinoflagellates and coccolithophores contribute to the phytoplankton biomass with the coccolithophores producing the most dramatic blooms. The dinoflagellates produce a red/orange (carotenoids), bioluminescent chlorophyll and the coccolithophores produce a milky white chlorophyll that increases the albedo of the ocean (*SEA2004F and SEA2005S Notes*) and contributes to the chalk cliffs of the south-west coast of England. This chalk contains the fossil remains of coccolithophores which indicates that this area was also rich in phytoplankton in the past. These cliffs extend all the way to the, more famous, White Cliffs of Dover on the south-east coast.



Figure 1: Hooken Cliffs, Devon
(Image Source: <https://ukfossils.co.uk/hooken-cliff/>)

The region of study chosen covers the coastlines of Cornwall, Devon and Dorset (about 350 km) and is situated between 51°N - 49.5°N and 6°W - 2°W which includes the South-Western English Channel and the Island of Guernsey.

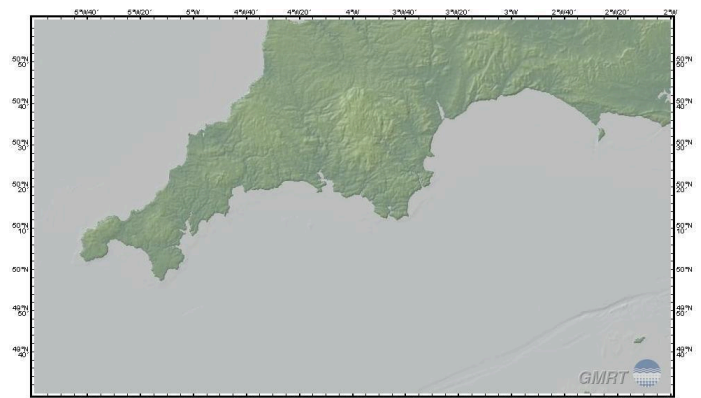


Figure 2: The South-Western English Channel
(Image Source: <https://www.gmrt.org/GMRTMapTool/>)

2. Data Sets and Descriptions with References

2.1. ESA-CCI Ocean Colour Data (Version 5.0)

Description: This data contains the climatological monthly means of satellite-derived chlorophyll concentration for the oceans with a spatial resolution of 4 km (high spatial resolution).

Purpose: Ocean colour and chlorophyll concentration can be used as a proxy for phytoplankton biomass to identify productivity or growth during phytoplankton blooms.

Format: NetCDF (Network Common Data Form)

Size: 1.79 GB

Temporal Scale: 1998-01-01 to 1997-12-01 with a resolution of P1M

Spatial Scale: global (90°N, 90°S, 180°W, 180°E) with a resolution of 4 km

Data Columns:

- chlor_a (float): the chlorophyll-A concentration measured in mg/m³
- time (datetime): the time over which the chlorophyll-A concentration was measured in months
- lon (float): the longitude measured in °E
- lat (float): the latitude measured in °W

Data Source: monthly composites of merged sensor products (sensors: MERIS, MODIS Aqua, SeaWiF, LAC & GAC, VIIRIS and OLCI)

Reference for ESA Ocean Colour CCI Data: Sathyendranath, S, Brewin, RJW, Brockmann, C, Brotas, V, Calton, B, Chuprin, A, Cipollini, P, Couto, AB, Dingle, J, Doerffer, R, Donlon, C, Dowell, M, Farman, A, Grant, M, Groom, S, Horseman, A, Jackson, T, Krasemann, H, Lavender, S, Martinez-Vicente, V, Mazeran, C, Mélin, F, Moore, TS, Müller, D, Regner, P, Roy, S, Steele, CJ, Steinmetz, F, Swinton, J, Taberner, M, Thompson, A, Valente, A, Zühlke, M, Brando, VE, Feng, H, Feldman, G, Franz, BA, Frouin, R, Gould, Jr., RW, Hooker, SB, Kahru, M, Kratzer, S, Mitchell, BG, Muller-Karger, F, Sosik, HM, Voss, KJ, Werdell, J, and Platt, T (2019) An ocean-colour time series for use in climate studies: the experience of the Ocean-Colour Climate Change Initiative (OC-CCI). *Sensors*: 19, 4285. doi:10.3390/s19194285

DOI for Version 5.0: Sathyendranath, S.; Jackson, T.; Brockmann, C.; Brotas, V.; Calton, B.; Chuprin, A.; Clements, O.; Cipollini, P.; Danne, O.; Dingle, J.; Donlon, C.; Grant, M.; Groom, S.; Krasemann, H.; Lavender, S.; Mazeran, C.; Mélin, F.; Müller, D.; Steinmetz, F.; Valente, A.; Zühlke, M.; Feldman, G.; Franz, B.; Frouin, R.; Werdell, J.; Platt, T. (2021): ESA Ocean Colour Climate Change Initiative (Ocean_Colour_cci): Version 5.0 Data. NERC EDS Centre for Environmental Data Analysis, 19 May 2021. doi:10.5285/1dbe7a109c0244aaad713e078fd3059a. <http://dx.doi.org/10.5285/1dbe7a109c0244aaad713e078fd3059a>

2.1. Global Multi-Resolution Topography (GMRT) Bathymetry Data (Version 4.4)

Description: This data has been downloaded from the GMRT MapTool (version 4.4) for the region of study. A low grid resolution has been used since the Ocean Colour Data has a resolution of 4 km. This data is unmasked and the grid has been filled with GEBCO 2025 Data where high resolution data does not exist for the GMRT Data. The following projection has been used to produce the GMRT Map: WGS84 Lat/Lon (EPSG: 4326).

Purpose: Bathymetry plays an important role in phytoplankton distribution so can be used to aid the identification of phytoplankton blooms.

Format: NetCDF (Network Common Data Form)

Size: 1.9 MB

Spatial Scale: regional (51°N, 49.5°N, 6°W, 2°W) with a resolution of 489 m/node

Data Columns:

- altitude (float): the height of the land and depth of the ocean measured in m (range: -176.2 m to 596.9 m)
- lon (float): the longitude measured in °E
- lat (float): the latitude measured in °W

Data Source: created with GMRT GridServer and extracted from GMRT Synthesis v4

Reference for GEBCO 2025 Bathymetry Data: GEBCO Compilation Group (2025) GEBCO 2025 Grid (doi:10.5285/37c52e96-24ea-67ce-e063-7086abc05f29)

Reference for GMRT Bathymetry Data: Ryan, W.B.F., S.M. Carbotte, J.O. Coplan, S. O'Hara, A. Melkonian, R. Arko, R.A. Weissel, V. Ferrini, A. Goodwillie, F. Nitsche, J. Bonczkowski, and R. Zemsky (2009), Global Multi-Resolution Topography synthesis, *Geochem. Geophys. Geosyst.*, 10, Q03014, doi: 10.1029/2008GC002332

3. Figures

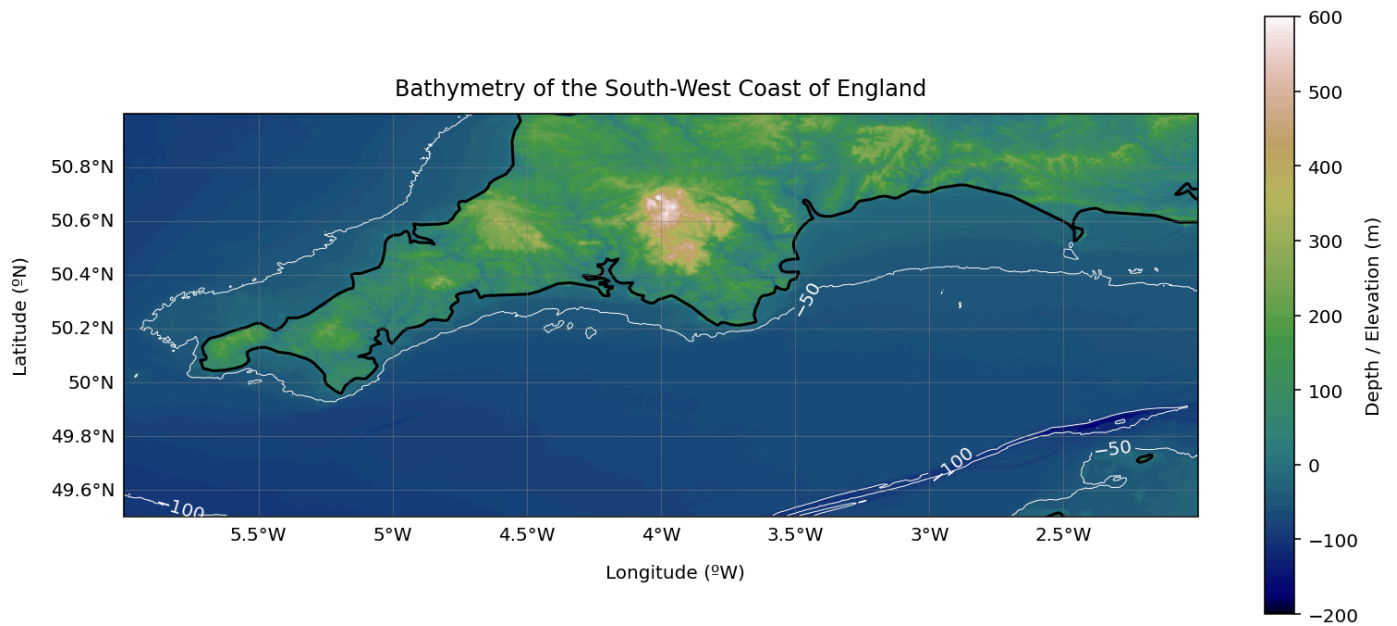


Figure 3: A bathymetry map showing the elevation along the South-West Coast of England and the depth of the South-Western English Channel with contours indicating 50 m and 100 m depth. The region of study contains the shallow sea that covers the continental shelf of Europe and contains a portion of Hurd Deep - an underwater valley that is the deepest part of the English Channel (indicated by depths greater than 100 m). (Data Source: GMRT Data)

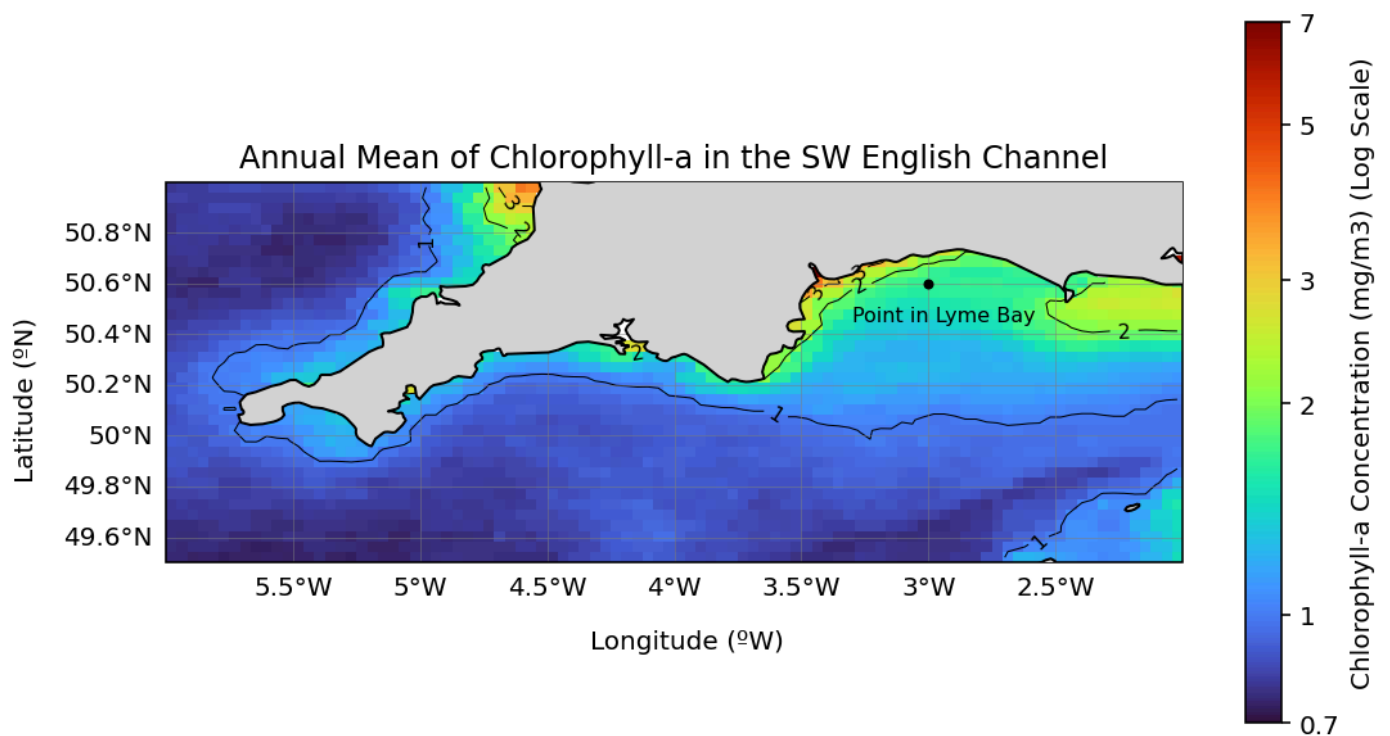


Figure 4: The annual mean of chlorophyll-a in the South-Western English Channel - as distance from the coastline increases and the water depth increases, the chlorophyll concentration decreases which is shown by the contours. The high concentration of chlorophyll at approximately 50.6°N and 3.5°W (chlorophyll-a > 3 mg/m³) is due to the River Exe which flows through the town of Exeter before reaching the English Channel at Exmouth, an estuary that is visible on the map. The river may carry pollutants and nutrients from the town and farms which increases the chlorophyll concentration in the estuary and along the coast. (Data Source: ESA Ocean Colour CCI Data)

Chlorophyll-a Monthly Means in the SW English Channel

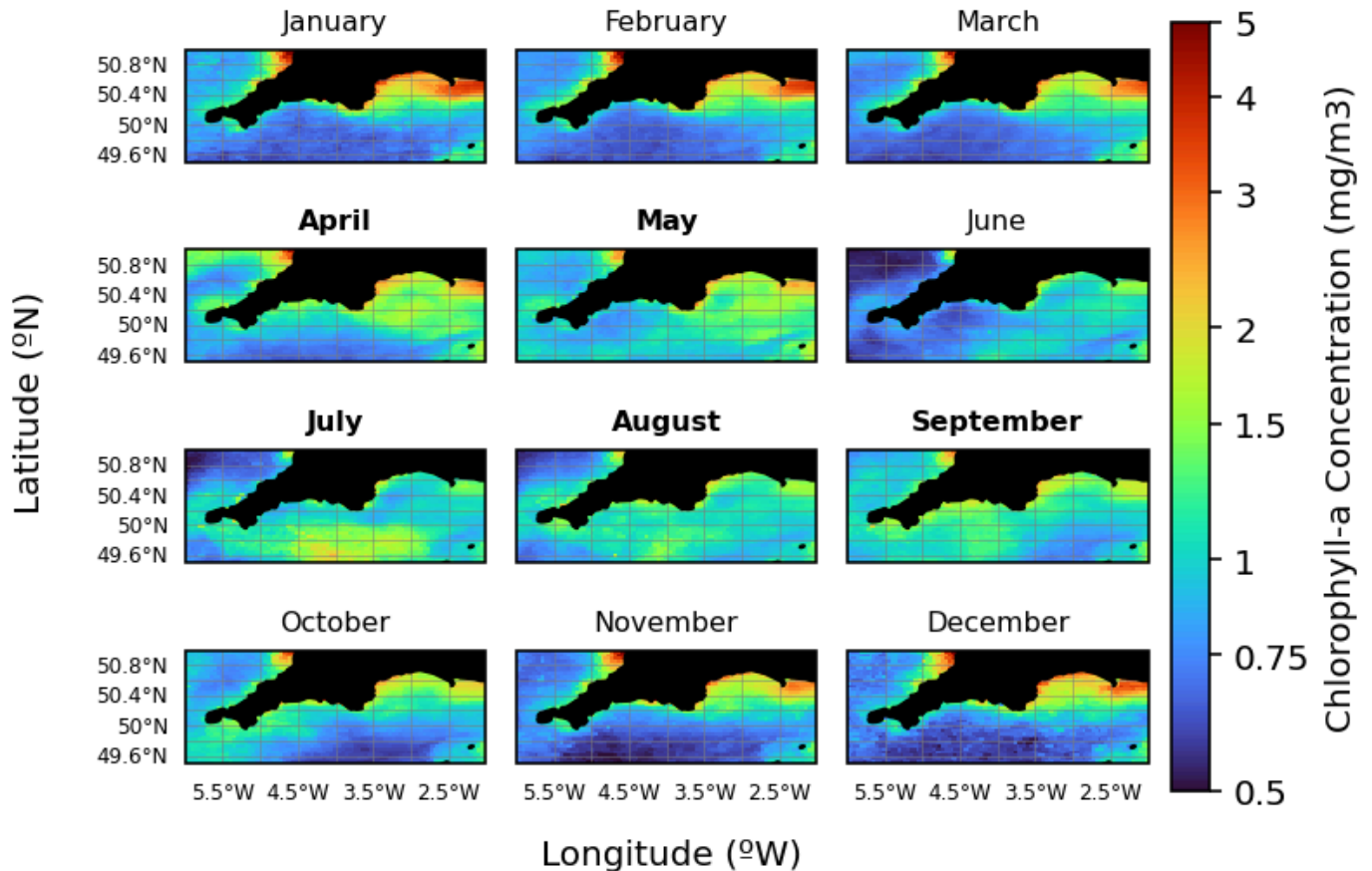


Figure 5: A faceted figure showing the monthly means of chlorophyll-a in the South-Western English Channel. Along the coast, the chlorophyll concentration is highest (red) during winter (December, January and February) due to increased river discharge into the ocean. The spring blooms are identified by the increase in chlorophyll concentration in the east (yellow/green) in April and May. A secondary bloom is also identified in the south-west in autumn (August and September). There is also an unusual summer bloom (July), indicated by the increased chlorophyll concentration in the south (orange/yellow). (Source: ESA Ocean Colour CCI Data)

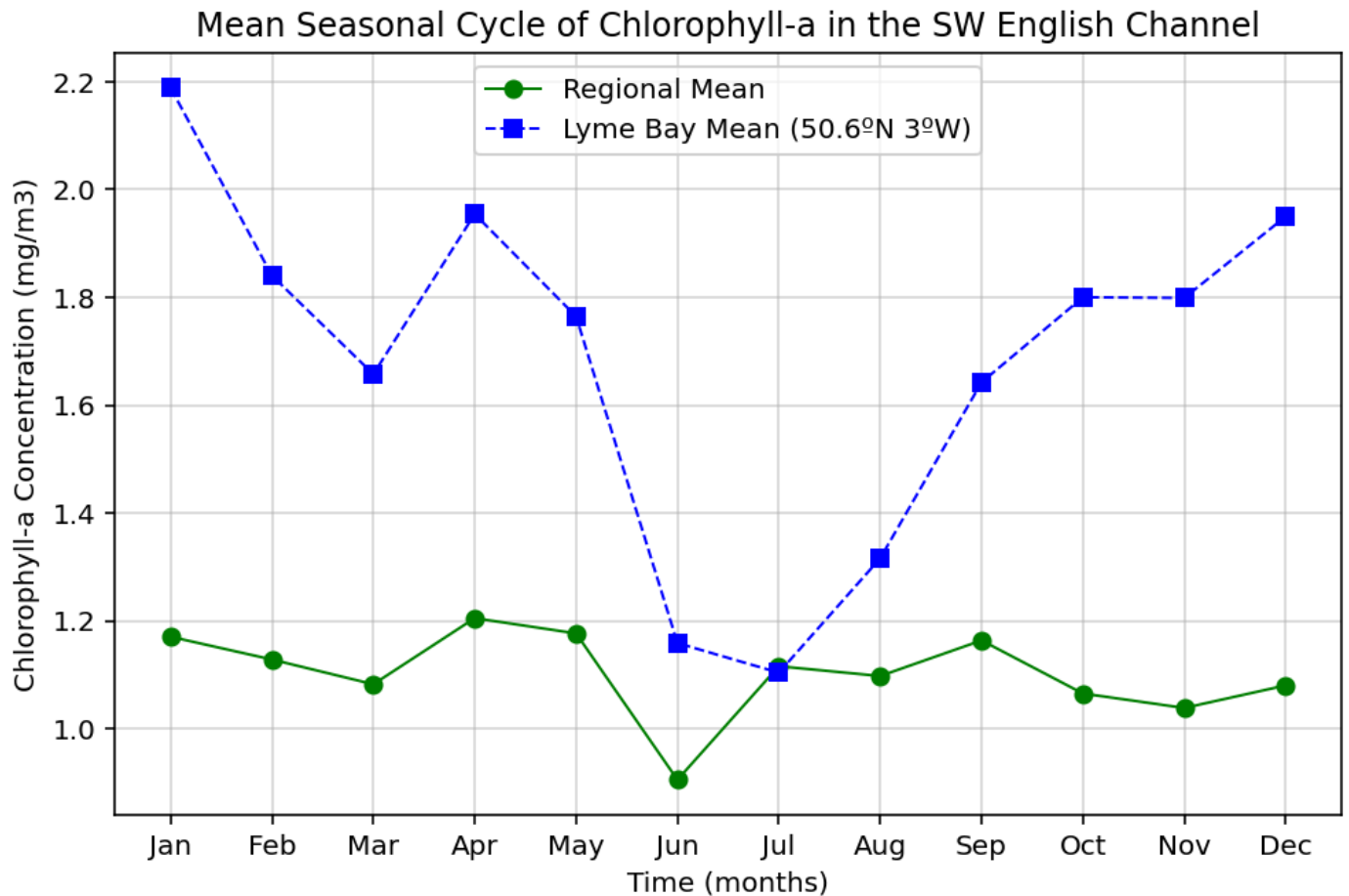


Figure 6: A timeseries comparing the mean seasonal cycle of the whole region and that of a point at 50.6°N and 3°W. This point is situated within Lyme Bay which has high concentrations of chlorophyll-a throughout the year. Both show a strong seasonal cycle with the spring bloom visible as an increase in chlorophyll in March and April. The timeseries for the point in Lyme Bay shows an increase in chlorophyll in autumn and winter (August to January) which may be due to river runoff and increasing storms. The two blooms are more pronounced in Lyme Bay than for the whole region. (Source: ESA Ocean Colour CCI Data)

4. Conclusion

By looking at the monthly chlorophyll maps and the timeseries, it is easy to identify seasonal changes in chlorophyll concentration that represent the seasonal blooms that are expected within the South-Western English Channel. These blooms occur in the transitional months with spring blooms (April and May) occurring as a result of increased sunlight and autumn blooms (August and September) occurring due to increased mixing by winds as the winter storms approach which is consistent with other observations and expectations. The change in chlorophyll can therefore be used as a proxy for phytoplankton biomass as ocean color changes occur at the same time as these blooms.

5. References

- GEBCO Compilation Group (2025) GEBCO 2025 Grid (doi:10.5285/37c52e96-24ea-67ce-e063-7086abc05f29)
- Google Maps and Google Earth (used for selecting the region of study and naming features within the region)
- Richard, C. (2022) Corio generation targets floating offshore wind in UK's Celtic Sea, Windpower Monthly. Available at: <https://www.windpowermonthly.com/article/1791900/corio-generation-targets-floating-offshore-wind-uks-celtic-sea> (Accessed: 25 March 2026). (map used for the naming of bathymetric features, bays and towns)
- Ryan, W.B.F., S.M. Carbotte, J.O. Coplan, S. O'Hara, A. Melkonian, R. Arko, R.A. Weissel, V. Ferrini, A. Goodwillie, F. Nitsche, J. Bonczkowski, and R. Zemsky (2009), Global Multi-Resolution Topography synthesis, *Geochem. Geophys. Geosyst.*, 10, Q03014, doi: 10.1029/2008GC002332
- Sathyendranath, S, Brewin, RJW, Brockmann, C, Brotas, V, Calton, B, Chuprin, A, Cipollini, P, Couto, AB, Dingle, J, Doerffer, R, Donlon, C, Dowell, M, Farman, A, Grant, M, Groom, S, Horseman, A, Jackson, T, Krasemann, H, Lavender, S, Martinez-Vicente, V, Mazeran, C, Mélin, F, Moore, TS, Müller, D, Regner, P, Roy, S, Steele, CJ, Steinmetz, F, Swinton, J, Taberner, M, Thompson, A, Valente, A, Zühlke, M, Brando, VE, Feng, H, Feldman, G, Franz, BA, Frouin, R, Gould, Jr., RW, Hooker, SB, Kahru, M, Kratzer, S, Mitchell, BG, Muller-Karger, F, Sosik, HM, Voss, KJ, Werdell, J, and Platt, T (2019) An ocean-colour time series for use in climate studies: the experience of the Ocean-Colour Climate Change Initiative (OC-CCI). *Sensors*: 19, 4285. doi:10.3390/s19194285
- Sathyendranath, S.; Jackson, T.; Brockmann, C.; Brotas, V.; Calton, B.; Chuprin, A.; Clements, O.; Cipollini, P.; Danne, O.; Dingle, J.; Donlon, C.; Grant, M.; Groom, S.; Krasemann, H.; Lavender, S.; Mazeran, C.; Mélin, F.; Müller, D.; Steinmetz, F.; Valente, A.; Zühlke, M.; Feldman, G.; Franz, B.; Frouin, R.; Werdell, J.; Platt, T. (2021): ESA Ocean Colour Climate Change Initiative (Ocean_Colour_cci): Version 5.0 Data. NERC EDS Centre for Environmental Data Analysis, 19 May 2021. doi:10.5285/1dbe7a109c0244aaad713e078fd3059a. <http://dx.doi.org/10.5285/1dbe7a109c0244aaad713e078fd3059a>
- SEA2004F and SEA2005S Lecture Notes (used for explanations regarding phytoplankton blooms)
- *Western English Channel Springs Into Life (2021) Plymouth Marine Laboratory*. Available at: <https://pml.ac.uk/news/western-english-channel-springs-into-life/#:~:text=The%20annual%20Spring%20bloom%20marks%20the%20end,often%20a%20signal%20of%20increased%20biological%20activity>. (Accessed: 16 March 2026).